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ORIGINAL ARTICLE

Machine Learning Approaches for Cardiovascular Disease Prediction: A Comparative Study

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Abstract

Cardiovascular disease (CVD) is a leading cause of death globally, particularly in South Asia, where high cholesterol intake contributes to its prevalence. This study aims to predict CVD using machine learning models applied to patient data from healthcare systems in Dhaka, Bangladesh. The study seeks to identify the most reliable model for early diagnosis and decision-making support. The dataset comprises 1019 patient records, collected from two prominent hospitals in Dhaka, and includes nine critical features. Several machine learning models were employed and rigorously tested using stratified fivefold cross-validation and best parameters were chosen using GridSearchCV. The model's performance was evaluated using metrics such as accuracy, precision, recall, F1-score, AUC and ROC curve analysis. To enhance interpretability, SHapley Additive exPlanations (SHAP) analysis was applied, focusing on global feature importance. Among the models tested, XGBoost exhibited the highest performance, achieving 97.12% training accuracy and 86.07% testing accuracy (AUC = 0.91). Random Forest also performed strongly with 83.08% testing accuracy and the highest AUC (0.92). Decision Tree and K-Nearest Neighbors achieved moderate results with testing accuracies of 78% and 74.63%, respectively. Logistic Regression and Support Vector Machine showed lower overall accuracy (~ 66%), though both attained high recall (0.91 and 0.95), indicating sensitivity to positive cases. These results highlight XGBoost's robustness while also demonstrating the trade-offs of alternative models. The results demonstrate that machine learning, particularly the XGBoost model with SHAP based explainability, offers a promising approach for diagnosing CVD with high accuracy. Incorporating this model into medical diagnostic systems can assist healthcare specialists in making more informed and accurate decisions, potentially reducing the morbidity and mortality associated with CVD.

Keywords Cardiovascular disease · Machine learning · Classification algorithms · Heart disease prediction · Health informatics

Abbreviations

CVD	Cardiovascular disease
SVM	Support vector machine
RF	Random forest
DT	Decision trees
LR	Logistic regression
KNN	K-nearest neighbor
ROC	Receiver operating characteristic
AUC	Area under ROC curve

Introduction

Cardiovascular disease (CVD) is a leading global health threat, causing approximately 17.9 million deaths in 2019, constituting 32% of global mortality, as reported by WHO [1]. In low- and middle-income nations like Bangladesh, rising CVD prevalence is linked to lifestyle changes, stress, and dietary habits. Non-communicable diseases (NCDs) now dominate Bangladesh's illness pattern, with CVD alone causing around 30% of total mortality [2, 3]. Early detection and timely intervention are therefore critical to reducing this burden. CVD encompasses various heart and vascular conditions, often preventable through lifestyle changes like

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quitting smoking, healthy eating, and regular exercise. However, early detection remains critical. Bangladesh, with over 165 million people and a 4.5–5.0% CVD prevalence [4–6], faces significant healthcare challenges, requiring efficient, cost-effective diagnostic methods.

Machine learning offers promising solutions, enabling accurate and reliable CVD prediction by analyzing data such as age, gender, blood pressure, cholesterol, and smoking habits. While much CVD research originates in well-funded countries, Bangladesh faces unique obstacles, including limited healthcare resources and data access.

Although prior studies have applied machine learning to CVD prediction, many use standardized or synthetic datasets and do not represent regional healthcare dynamics. To address this gap, the present study uses a real-world dataset collected from Bangladeshi hospitals, an underrepresented region in medical ML research. Unlike many prior works that rely on standard benchmark datasets, this dataset reflects region-specific risk factors, healthcare challenges, and lifestyle patterns. Additionally, it evaluates six machine learning classifiers using both stratified fivefold cross-validation and hold-out validation, followed by SHAP-based interpretability analysis. Both predictive accuracy and interpretability analysis were employed to provide clinical transparency by explaining how features such as age, blood pressure, cholesterol, and lifestyle behaviors influence predictions. This integration of a unique regional dataset, rigorous evaluation strategies, and interpretable machine learning contributes a novel perspective to CVD prediction research, with the potential to inform practical diagnostic support tools in resource-limited healthcare systems.

In this research, machine learning has been used to predict CVD based on different types of data, such as age, gender, blood pressure, cholesterol, glucose level, physical activity, smoking, etc. The best model has been found based on the data to predict whether the patient have CVD or not. After that testing and evaluating was done on the models. This study focuses on enhancing CVD prediction in a developing nation like Bangladesh. The major goal is to evaluate the accuracy of predicting patient diseases in a context influenced by a variety of socioeconomic factors and healthcare problems. The findings of this study have the potential to inform more focused and efficient healthcare interventions in similar developing countries, therefore contributing to the global advancement of predictive healthcare models.

Literature Review

Extensive research has been conducted to predict cardiovascular diseases (CVD) and identify critical risk factors, with machine learning techniques playing a vital role. Researchers have explored various machine learning methods using global datasets to enhance early detection and prevention of CVD. However, many rely on publicly available datasets like Kaggle, which may not fully represent real-world patient data. Utilizing location-specific healthcare data can improve predictive accuracy and provide deeper insights into the manifestation of CVD.

Weicheng et al. in 2021, used a dataset of 70,000 samples to predict CVD with machine learning methods like SVM, RF, and LR. SVM achieved the highest accuracy (71.48%) and AUC (78.84%) using correlation coefficient methods for feature extraction [7]. Syed et al. in 2020, applied deep learning and machine learning techniques such as ANN, KNN, and SVM, with ANN achieving 85.24% accuracy on Kaggle data [8]. Atharv et al. in 2020, emphasized BMI as a feature, using various algorithms, with Decision Trees performing best (73.13% testing accuracy) [9].

Ibashisha et al., enhanced accuracy by integrating features like blood pressure and BMI, finding KNN to perform best among classifiers like LDA, NB, DT, and RF [10]. Vansh et al. created an iOS app for CVD prediction, achieving 72.70% accuracy with XGBoost [11]. Abdul et al. tested nine classifiers and identified SVM as the most effective, but they noted inefficiencies in handling large datasets and challenges with hyperparameter tuning, which impacted performance and generalizability [12].

Arsalan et al., gathered data from two major teaching hospitals in Pakistan, employing algorithms such as SVM, LR, DT, RF, and NB. RF achieved the highest accuracy (85.01%), sensitivity (92.11%), and AUC (87.73%) [13]. Tamanna et al. in 2021, used the XGBoost algorithm, achieving 75.10% accuracy, and highlighted the role of feature selection using RF [14]. Khairul et al. in 2022, used four algorithms (RF, KNN, XGBoost, DT) and found accuracy improved from 73.72 to 81.14% when reducing the dataset [15]. Recently, Donmez et al. [16] conducted a study to demystify hypertension risk prediction (623 patients) using Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP). The XGBoost model demonstrated exemplary performance, achieving an accuracy of 99.4%, precision of 100%, recall of 97.3%, and an F1-score of 98.6%. Additionally, the same author introduced a quantum-enhanced machine learning model specifically designed for hypertension prediction, integrating quantum feature transformation with classical algorithms to deliver precise and reliable results. The hybrid XGBoost–quantum model demonstrated high performance in 2025 [17], achieving an accuracy of 98.40%, precision of 99.3%, recall of 98.6%, F1-score of 98.9%, and an AUC-ROC score of 0.99.

Mohammad et al. in 2021, conducted field research in Sylhet, Bangladesh, collecting data from hospitals and institutions. Their SVM model achieved 91% accuracy with optimized feature selection [18]. Md et al. in 2022, used

the mean imputation method and a reduced dataset of 10 features to achieve 95.63% accuracy with RF [19]. Shahiduzzaman et al. in 2022, tested ensemble classifiers, with KNN achieving 75% accuracy, outperforming RF, Bernoulli NB, Gradient Boosting, and LR [20].

Despite the challenges in South Asia, including limited datasets and infrastructure, researchers in Bangladesh are committed to advancing CVD prediction through innovative machine learning techniques, showcasing the potential for significant healthcare improvements.

Methodology

The methodology for cardiovascular disease prediction is summarized in Fig. 1, illustrating steps from data collection and preprocessing to model training and testing. Duplicate data were removed, and the dataset was split (80% training, 20% testing). Multiple classifiers, including XGBoost and Logistic Regression, were applied, with stratified fivefold cross-validation ensuring robust evaluation.

Data Collection

Unlike widely used public datasets such as Framingham or UCI, this study employs a local hospital dataset from Bangladesh, which reflects region-specific risk factors, healthcare limitations, and lifestyle differences. Such datasets remain underexplored in current CVD prediction literature, adding unique value to this work. Data for this study was collected from two prominent hospitals in Dhaka, Bangladesh: the National Institute of Cardiovascular Diseases and the Ibrahim Cardiac Hospital and Research Institute. Like previous studies such as Mohammad et al. [18], the data collection process adhered to local hospital protocols due to the sensitivity of medical records and patient privacy concerns. The dataset comprises 1019 records with 8 attributes and 1 binary target column (0 = no cardiovascular disease, 1 = presence of cardiovascular disease). The primary goal is to build an accurate model capable of predicting the presence of cardiovascular disease based on the provided features. Table 1 provides details of the dataset's attributes and their descriptions.

Data Pre-processing

After collecting data, preprocessing was essential to clean invalid, missing, or noisy information that could impair machine learning model's performance. The initial dataset contained 1019 rows and 9 columns, which were checked for duplicates. The target variable, cardiovascular disease, was moderately imbalanced, with 662 instances (64.97%) labeled as CVD-positive and 357 instances (35.03%) labeled as CVD-negative. No explicit oversampling or weighting techniques were applied. Instead, we relied on stratified fivefold cross-validation, which preserved class proportions across folds and reduced bias toward the majority class during evaluation. Model optimization was performed using

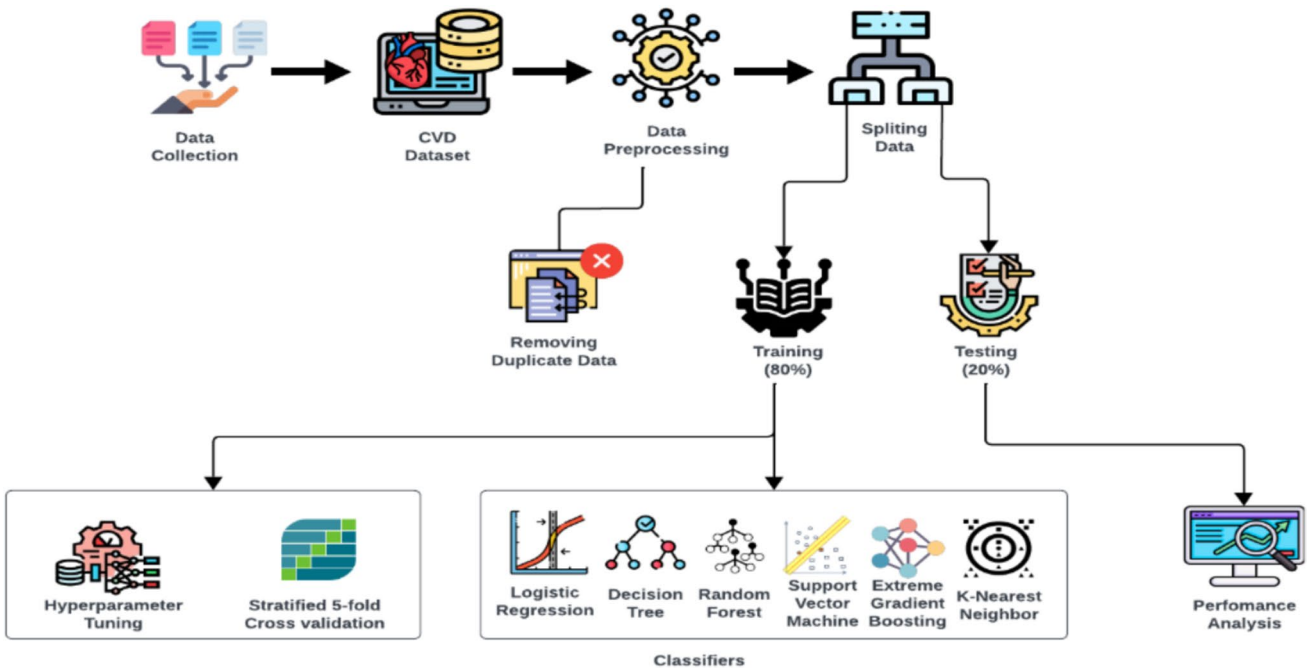


Fig. 1 Work-flow diagram of the cardiovascular disease prediction

Table 1 Description of attributes in the collected dataset, including data types and explanatory information

Attribute/feature	Value type	Variable	Variable type	Explanation
Age	Int	Age	Objective	Age in years
Gender	Binary	Gender	Objective	Female = 1, Male = 2
Systolic blood pressure	Int	SystolicBP	Examination	Blood Pressure
Diastolic blood pressure	Int	DiastolicBP	Examination	Blood Pressure
Cholesterol level	Categorical code	CholesterolLevel	Examination	1: Normal (< 200 mg/dL), 2: Moderate (200–239 mg/dL), 3: High Risk (≥ 240 mg/dL)
Glucose level	Categorical code	GlucoseLevel	Examination	1: Normal (3.5–6.0), 2: Prediabetic (6.1–6.9), 3: Diabetic (≥ 7)
Smoking	Binary	Smoking	Subjective	0: Non-Smoker, 1: Smoker
Physical activity	Binary	Activity	Subjective	0: Not Active, 1: Active
Cardiovascular disease	Binary	Disease	Target variable	0: Negative, 1: Positive

F1-score in GridSearchCV to balance sensitivity and precision. Additionally, ensemble tree-based classifiers such as Random Forest and XGBoost are relatively robust to moderate imbalance, further justifying the decision not to apply synthetic resampling methods such as synthetic minority oversampling technique (SMOTE). After removing redundant records, 1001 unique patient records were retained. A review confirmed no missing values, indicating successful cleaning. For the binary classification task (0 = no cardiovascular disease, 1 = presence of cardiovascular disease), Pandas was used for data manipulation, and scikit-learn's StandardScaler scaled the data to ensure better model performance during training.

Data Description/Exploratory Data Analysis

As previously mentioned, the dataset for this paper comprises 8 attributes and 1 target column. These attributes include Age, Gender, Systolic Blood Pressure, Diastolic Blood Pressure, Cholesterol Level, Glucose Level, Smoking, and Physical Activity, with the target column being cardiovascular disease. In this step, the paper has systematically examined the individual impacts of each attribute within the dataset.

Age

The Age attribute represents the age of each participant in the study. It is a continuous variable indicating the years of age for each individual. From Fig. 2 it can be seen that the disease increases as the age increases.

Gender

Gender is a binary variable where 1 represents female and 2 represents male. As shown in Fig. 3, the records for men and women are nearly identical, indicating that the predictions will be free from gender bias. This balance is crucial

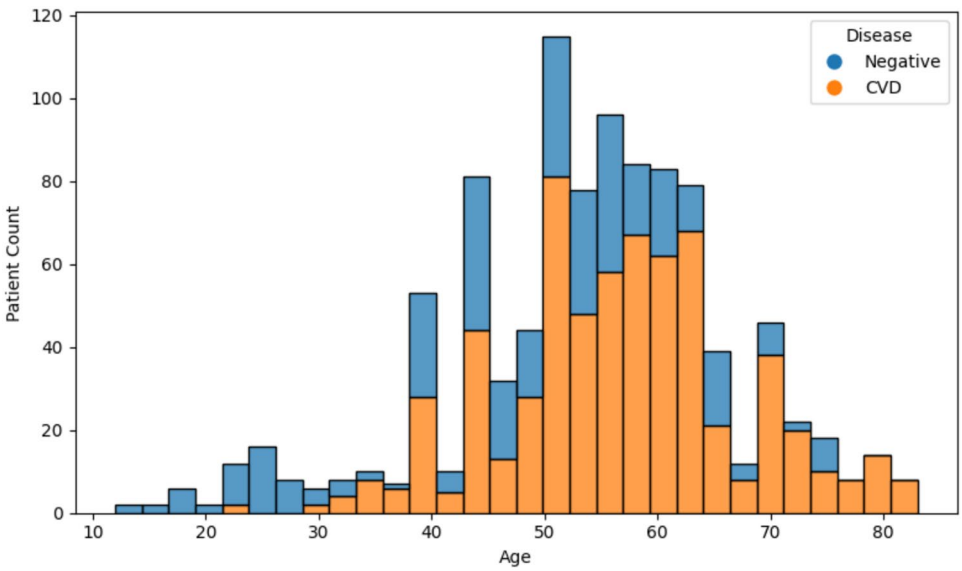


Fig. 2 Patient age distribution, broken down by disease status

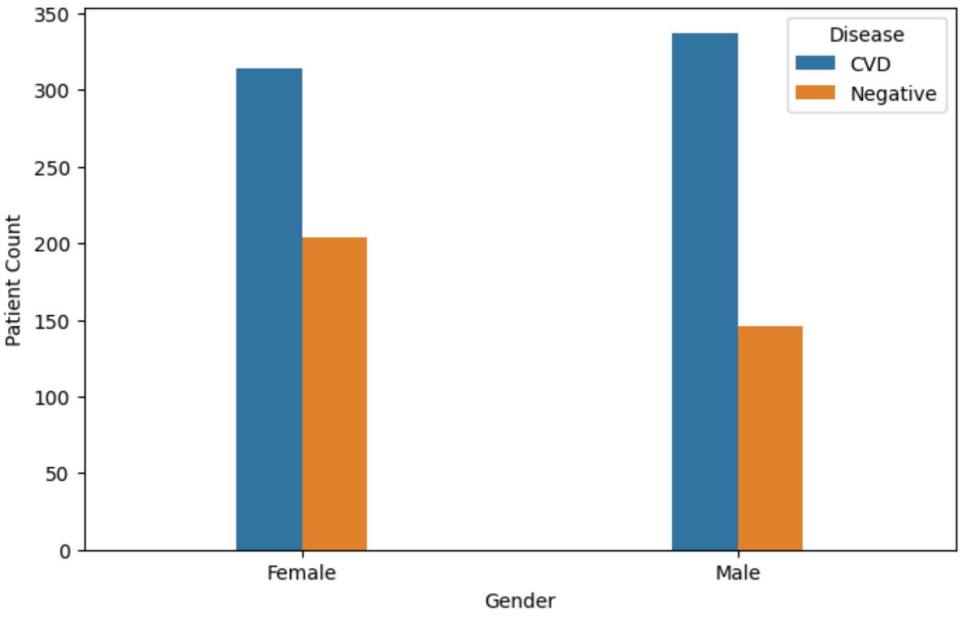


Fig. 3 A count map of patient gender distribution by cardiovascular disease (CVD) status shows the prevalence of CVD across genders and suggests gender-specific risk factors and therapies

for machine learning model development, as training on a dataset with gender imbalances could reduce the model's performance on more evenly distributed data.

Systolic and Diastolic Blood Pressure

Systolic Blood Pressure measures arterial pressure during heartbeats, while Diastolic Blood Pressure reflects arterial pressure during rest. Both play significant roles in cardiovascular health. Blood pressure is categorized as healthy ($< 120/ < 80$ mmHg), elevated (120–129/ < 80 mmHg), Stage 1 hypertension (130–139/80–89 mmHg), Stage 2 hypertension ($\geq 140/ \geq 90$ mmHg), and hypertensive crisis ($> 180/ > 120$ mmHg). Instead of categorizing, continuous values for Systolic and Diastolic Blood Pressure were used for analysis. Figures 4 and 5 illustrate their impacts on cardiovascular disease.

Cholesterol Level

Figure 6 illustrates the correlation between increasing cholesterol levels and heightened CVD risk. It's noteworthy that South Asian dietary patterns, often high in cholesterol-rich foods, contribute to elevated cardiovascular risks, underscoring the importance of addressing cholesterol levels in healthcare interventions.

Glucose Level

Diabetes is highly prevalent in South Asian countries, including Bangladesh, due to genetic predisposition and lifestyle factors. As shown in Fig. 7, diabetic patients have the highest occurrence of cardiovascular disease (CVD) compared to prediabetic and normal individuals, emphasizing the strong link between glucose levels and cardiovascular

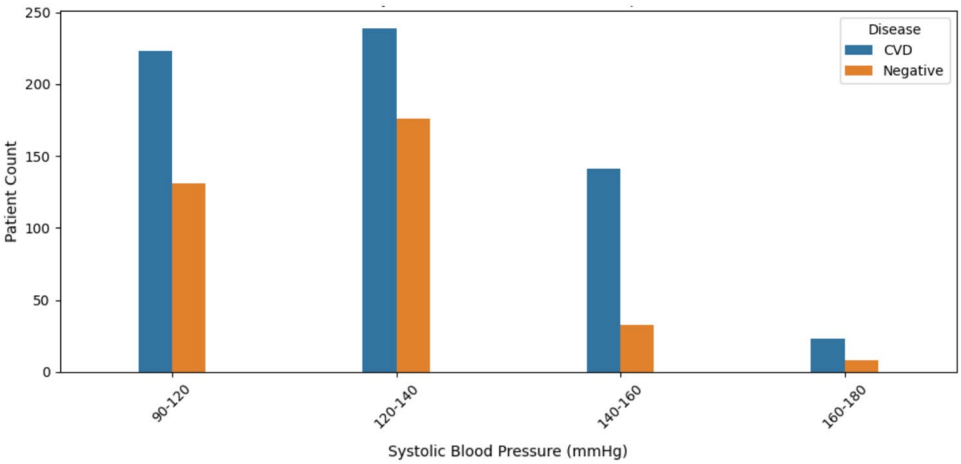


Fig. 4 Count plots of patient's systolic blood pressure values by illness state may help identify risk factors and advise cardiovascular health interventions

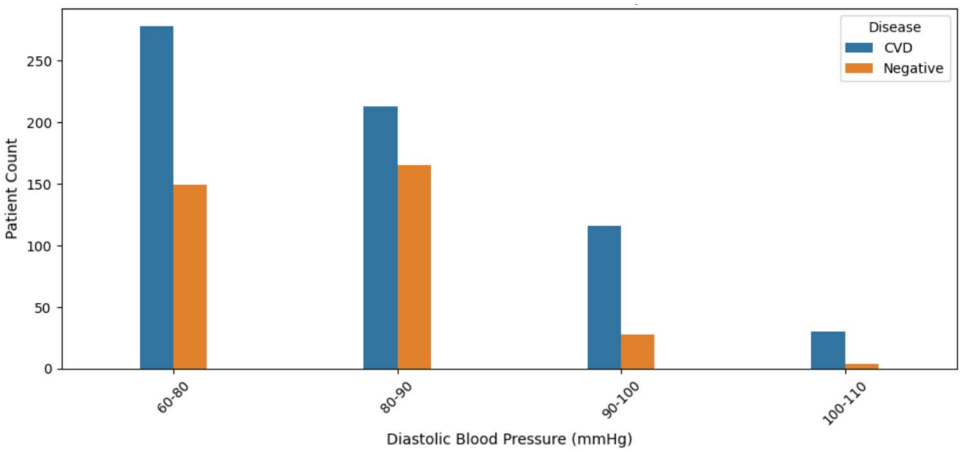


Fig. 5 Count plots of diastolic blood pressure levels by illness state may help identify risk factors and propose cardiovascular health interventions

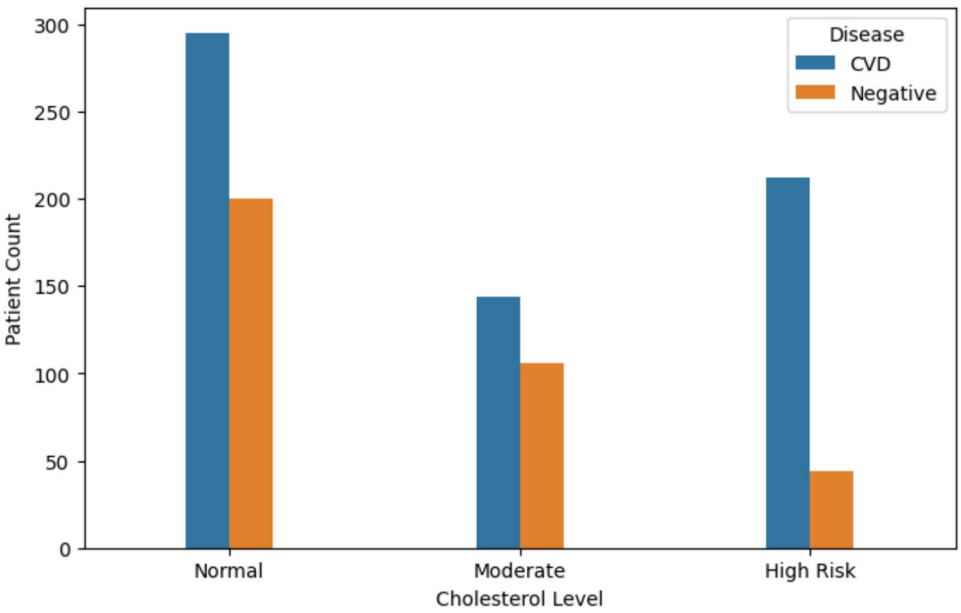


Fig. 6 Analysis of patient cholesterol levels by disease status: normal (category 1), moderate (category 2), and high risk (category 3)

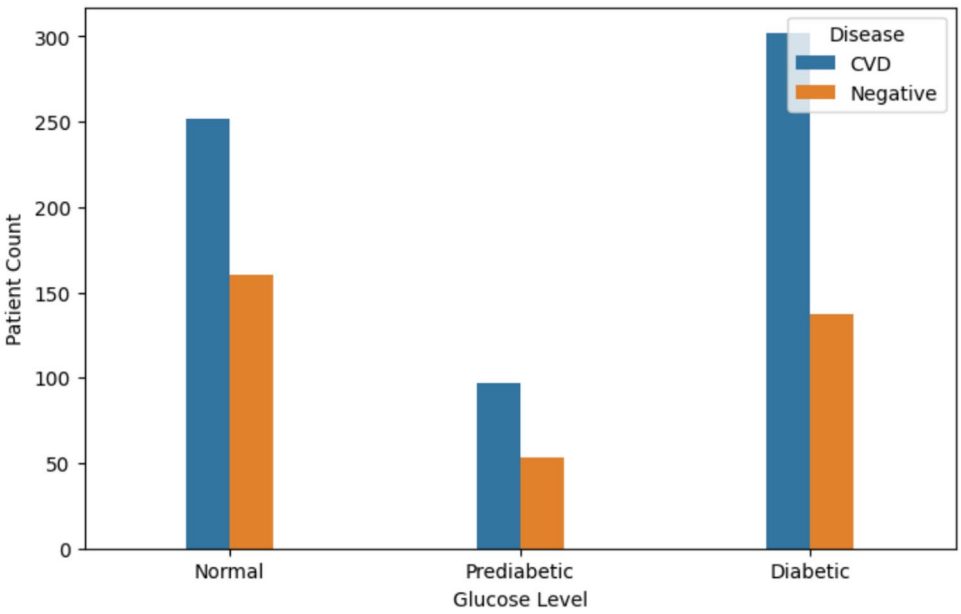


Fig. 7 Comparing normal, prediabetic, and diabetic glucose levels in cardiovascular disease patients

risk. Sedentary lifestyles and diets rich in refined sugars and carbohydrates further exacerbate this issue. Addressing these factors is essential for developing effective healthcare strategies and interventions for at-risk populations.

Smoking

Figure 8 shows a higher smoking prevalence among individuals with cardiovascular disease in Bangladesh, highlighting a strong association between smoking and the risk of developing cardiovascular disease. This underscores the importance of considering behavioral factors like smoking habits when analyzing and predicting cardiovascular risks, especially in regions like Bangladesh, where smoking rates are significantly high.

Physical Activity

Like smoking it is also a behavioral attribute which is binary. If a person is physically not active the value is 0, whereas a value of 1 indicates that they are. Figure 9 also shows that physical activity has an impact on CVD.

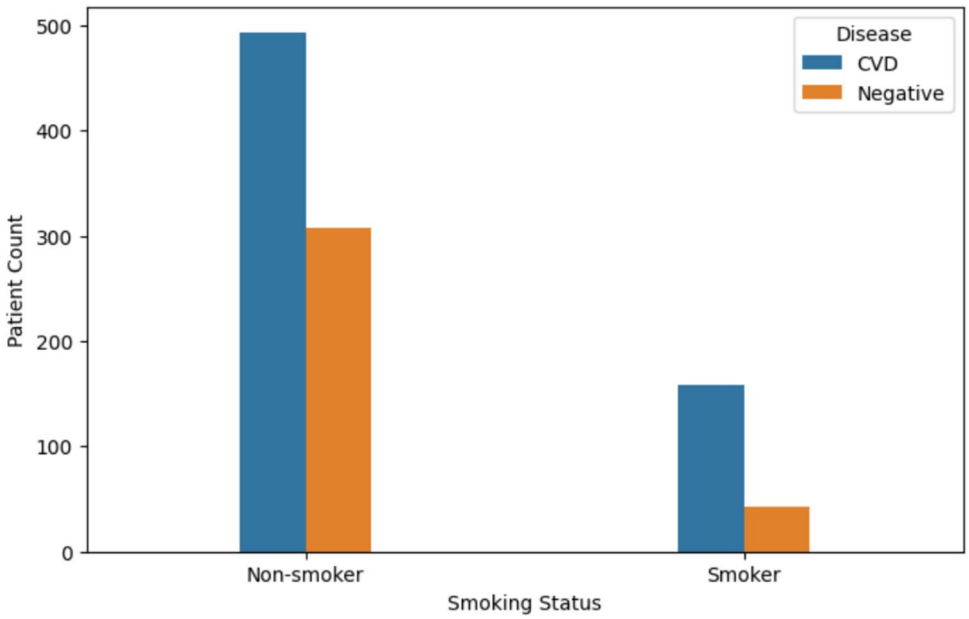


Fig. 8 Examining smoking and disease: patient smoking habits by disease status

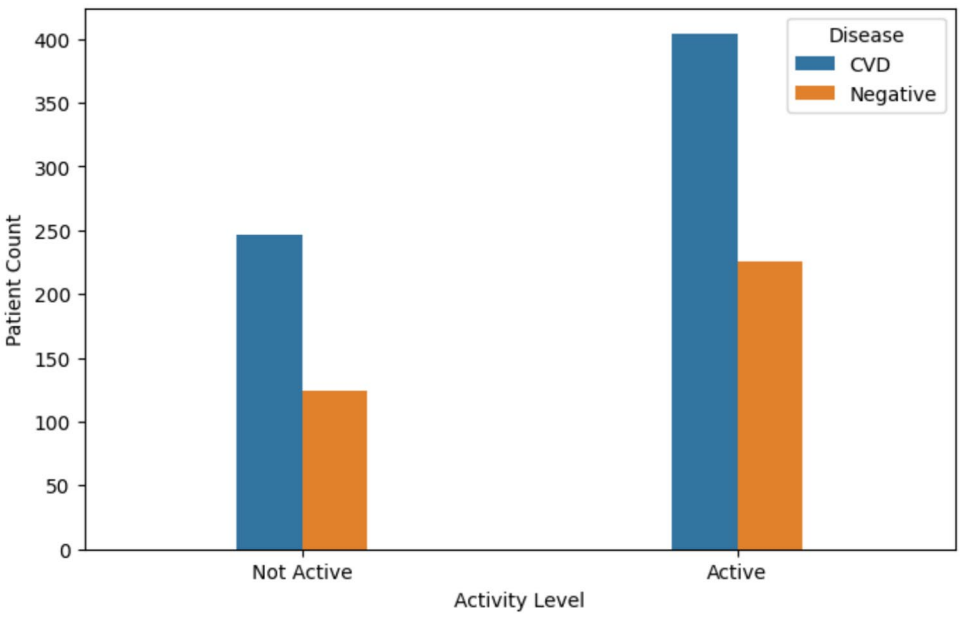


Fig. 9 Count plot of physical activity among patients stratified by disease status

Cardiovascular Disease

This column is the target column for determining whether or not a person has cardiovascular disease shown in Fig. 10. The two values are 0 and 1. A score of 0 means the person doesn't have heart disease, while a score of 1 means they do.

Feature Selection

Feature selection is vital for enhancing accuracy and performance by identifying essential attributes for predicting cardiovascular disease. The correlation coefficient matrix was used to evaluate the linear relationships between variables, as shown in Fig. 11. The heatmap highlights the strongest positive correlation with age (0.31), indicating increased CVD risk with age. Moderately positive correlations were found for smoking (0.14) and cholesterol level (0.18). Conversely, physical activity showed a slight negative correlation (− 0.02), suggesting it may slightly reduce CVD risk.

This study incorporated all the attributes in the cardiovascular disease prediction model even after noticing low scores of some attributes. This choice was made since it is recognized that every attribute adds to the total complexity of predicting CVD, and that leaving out any component may result in the loss of small patterns or possible interdependencies. Even though some attributes individually have less predictive value, this approach promotes a thorough analysis, taking into account all available information to ensure an accurate evaluation of potential factors for cardiovascular disease.

In addition, pairwise correlation only measures linear associations and does not capture non-linear or interaction effects that may become relevant when features are combined in machine learning models. For example, physical activity, although weakly correlated on its own, can influence predictions when considered with age, cholesterol, or smoking status. SHAP analysis further confirmed that even such low-correlation features contributed marginally to model decisions, aligning with established clinical knowledge that lifestyle behaviors play a role in cardiovascular health. For these reasons, all features were retained to maximize both statistical and clinical interpretability.

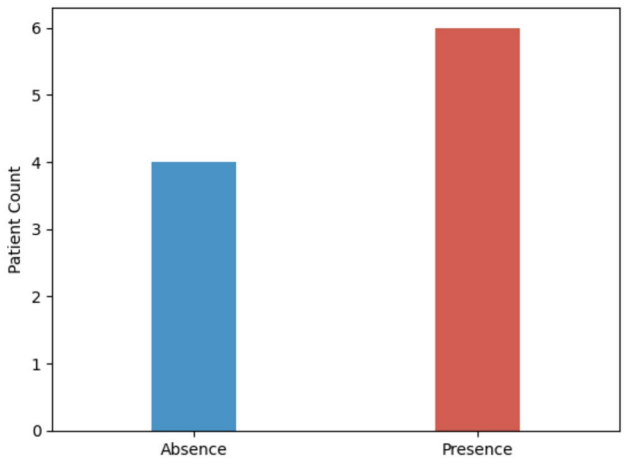


Fig. 10 Statistical analysis of the disease status of individual patients

Data Splitting

Data splitting is crucial in machine learning. It involves dividing a dataset into two parts: a training set and a testing set. The training set is used to teach the model to recognize patterns, while the testing set is used to check how well the model performs on new data it hasn't seen before. In this study, 80% of the cardiovascular disease dataset was used

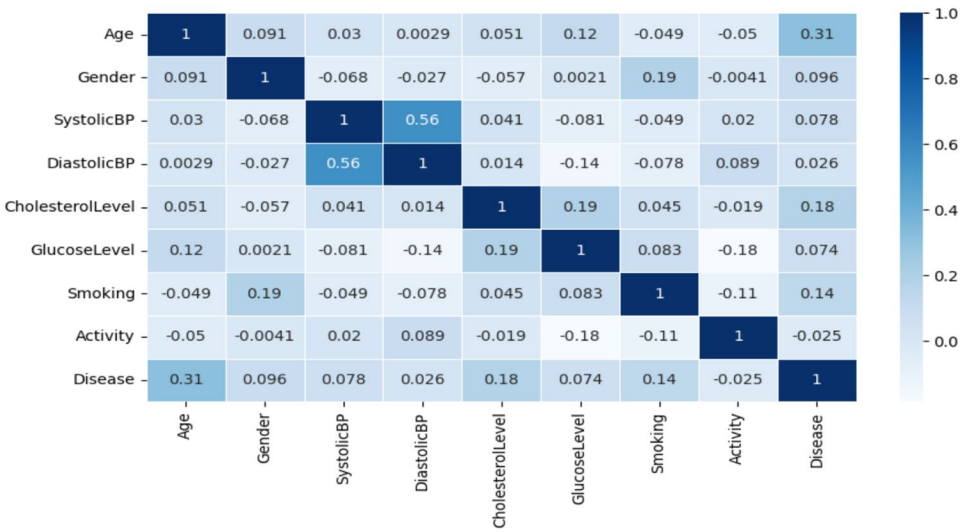


Fig. 11 An illustration of the correlations between all of the features of the dataset using a heatmap

for training, and 20% was used for testing. This allows for evaluating the model's performance accurately.

Training Model/Classifiers

Supervised classification machine learning methods were employed in this research because classification algorithms are suitable when the output variable is binary, indicating two classes, such as Yes–No, Male–Female, True–False, and so on. The objective of this model is to predict whether a patient has cardiovascular disease or not, making the output variable categorical. Consequently, six classification algorithms were utilized to predict cardiovascular disease in this study. The experiments were conducted using the scikit-learn Python library. Following the removal of duplicate rows, Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB), and K-Nearest Neighbor (KNN) were applied to the dataset containing 1001 samples. For interpretability, we employed SHAP (SHapley Additive Explanations) on the best performing XGBoost model. In this study, we restricted our analysis to global SHAP values, which summarize the average influence of features across the dataset.

Logistic Regression

Logistic Regression is a supervised learning classification algorithm that was used to predict the probability of cardiovascular disease. A logistic model can be used to predict whether a person has CVD or not by calculating the weighted sum of input variables such as age, systolic blood pressure, cholesterol level, smoking, and so on. The sigmoid function transforms this sum into a probability score between 0 and 1, indicating the likelihood of having CVD. Sigmoid/ Logistic function is described by the following formula [21]:

$$g(z) = \frac{1}{1 + e^{-z}} \tag{1}$$

The output of the sigmoid function must be between 0 and 1. If the probability is above a threshold (usually consider 0.5), then it is classified as cardiac (1). Otherwise, it is classified as non-cardiac (0).

Support Vector Machine

Support Vector Machine (SVM) employs a hyperplane as a decision boundary to separate classes for CVD prediction. It generates an optimal boundary to classify instances based on attributes like age, gender, blood pressure, cholesterol, and physical activity. SVM prioritizes critical cases by selecting support vectors-extreme points that influence the decision boundary. The distance between the hyperplane and these support vectors impacts the prediction. As seen in Table 2, SVM performed better on training data, reducing overfitting and demonstrating its effectiveness in capturing complex relationships in patient data.

Decision Tree

Decision Tree predicts CVD using a hierarchical structure. Key attributes like age, systolic and diastolic blood pressure, cholesterol, diabetes, and smoking form the basis for splitting at the root node. Internal nodes evaluate specific conditions, while leaf nodes store the final CVD prediction as positive or negative. Predictions follow explicit decision rules from root to leaf, with binary conditions for categorical features and thresholds for numerical ones. This interpretable model provides insights into factors influencing CVD, such as predicting CVD as negative if age and systolic blood pressure fall within specific ranges, or positive otherwise.

Random Forest

Random forest is a classification method based on Decision Trees. It creates many trees forming a forest. Each tree predicts CVD independently. The final prediction is based on the majority vote of all trees. Whichever class label gets the most votes is considered the prediction for the test data. Every single piece of data in the dataset goes through this process once again. The ability of Random Forest to produce several Decision Trees in the prediction of cardiovascular disease offers a thorough examination of numerous features, enabling the model to capture complex interactions within the data. Random Forest uses a majority vote process to combine individual forecasts into a final prediction that is more robust and dependable for every test data point.

XGB (Extreme Gradient Boosting)

Extreme Gradient Boosting (XGBoost) is an ensemble learning method for predicting CVD. It combines predictions from multiple weak models, typically decision trees, to enhance accuracy and reliability. The process begins with an initial model, and subsequent models iteratively focus on minimizing residuals (differences between predicted and actual values). Each new model is trained to correct errors from previous models, and their predictions are combined and weighted to update the overall forecast. This iterative refinement continues, with an emphasis on addressing prior mistakes. XGBoost's ability to handle imbalanced data, integrate regularization techniques like L1 and L2, and deliver high performance makes it an excellent choice for scalable and effective CVD prediction.

Table 2 Performance metrics for six machine learning models after hyper-parameter tuning

No	Model	Best parameters	Parameter range tested	Train accuracy (%)	Test accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	AUC (%)
1	XGBoost	learning_rate = 0.1, max_depth = 5, n_estimators = 200	n_estimators ∈ {100,200,300}, max_depth ∈ {3,5,7}, learning_rate ∈ {0.01,0.1,0.2}	97.12	86.07	89.08	87.60	88.33	91
2	Random Forest	max_depth = 10, n_estimators = 200	n_estimators ∈ {100,200,300,500}, max_depth ∈ {5,10,15,None}	97.02	83.08	82.71	90.10	86.61	92
3	Decision Tree	max_depth = 10	max_depth ∈ {3,5,10,None}	91.50	78.11	80.31	84.30	82.26	84
4	K-Nearest Neighbors	n_neighbors = 5	n_neighbors ∈ {3,5,10,15,18,20}	83.87	74.63	76.92	82.64	79.68	83
5	Logistic Regression	C = 0.1	C ∈ {0.01,0.1,1,10}	70.37	66.67	66.07	91.74	76.82	71
6	Support Vector Machine	C = 0.1	C ∈ {0.01,0.1,1,10}, kernel ∈ {linear,rbf}	71.63	66.17	64.80	95.87	77.33	71

KNN (K-Nearest Neighbor)

Another useful technique in cardiovascular disease prediction models is K-Nearest Neighbors (KNN). Rather than learning immediately from the training set, it stores the dataset and processes it later during classification. In CVD prediction, this algorithm efficiently manages data. As new data arrives, KNN categorizes it into a subset similar to the already stored data. The classification of a data point in CVD prediction is determined by a majority vote from its closest neighbors. The selection of the distance measure and the number of neighbors (k) significantly impacts KNN's performance. In the proposed approach, elbow method has been employed to determine the optimal number of neighbors, setting it to $k = 18$ for effective CVD prediction.

Hyperparameter Tuning Strategy

To ensure optimal model performance and avoid overfitting, hyperparameter tuning was performed using GridSearchCV with fivefold stratified cross-validation. For KNN, the elbow method suggested $k = 18$. For Random Forest, the number of trees was varied from 100 to 500, and 300 yielded the best performance. XGBoost parameters such as max_depth, learning_rate, and n_estimators were optimized using a grid search over values (max_depth = [3, 5, 7], learning_rate = [0.01, 0.1], n_estimators = [100, 200, 300]). The best configuration was then applied to the final hold-out set for evaluation.

Evaluation Model

When evaluating the effectiveness of machine learning models, evaluation metrics are crucial techniques. The model's performance has been assessed using f1, accuracy, precision, and recall [22]. They are essential for determining how well the model is working and whether it satisfies the objectives and specifications of the goal. These matrices are described below

- Precision

  Precision is the proportion of correctly predicted positive cases out of all predicted positive cases. In CVD, high precision means a low chance of false positives and reliable identification of true positive cases.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

- **Recall**

  Sensitivity, also called recall, evaluates how effectively the model detects all actual positive cases of cardiovascular disease (CVD). It is the proportion of all actual positive CVD cases to all accurately anticipated positive CVD cases.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

- **F1 Score**

  The F1 score, which is the harmonic mean of recall and precision, provides a balanced assessment, especially useful when dealing with imbalanced class sizes in CVD prediction.

$$\text{F1 Score} = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{4}$$

- **Accuracy**

  Accuracy measures how well the model's predictions match the actual outcomes. It's the ratio of correctly predicted observations to all observations, including both true positives and true negatives.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{5}$$

  where, $TP = TruePosition$ (Total number of patients correctly identified that they have CVD). $FP = FalseNegative$ (Total number of patients incorrectly identified that they have CVD). $TN = TrueNegative$ (Total number of patients who correctly identified that they have no CVD). $FN = FalseNegative$ (Total number of patients incorrectly identified that they have no CVD).

AUC-ROC Curve

The ROC curve is a common statistical tool used for data classification. It shows how the model makes decisions at different confidence levels. It consists of two lines: one shows how often the model correctly identifies positive cases, while the other shows how often it incorrectly identifies negative cases as positive. This graph helps us understand how well the model performs, which is crucial for predicting cardiovascular disease accurately. The area under the ROC curve (AUC) indicates the classifier's ability to distinguish between different classes. It shows how well the model can tell apart positive and negative categories.

Result and Discussion

The experiments were conducted using Google Colab on an 11th-generation Intel Core i7 system with 16 GB RAM. To ensure robust evaluation and minimize overfitting, Stratified fivefold Cross-Validation was applied during hyperparameter tuning. A pipeline-based implementation was used, combining standard scaling with GridSearchCV for parameter optimization. Model performance was assessed using metrics such as accuracy, precision, recall, F1 score, and AUC on the 20% test data.

Significant gaps between training and testing accuracies were observed for some models. For instance, Random Forest achieved a training accuracy of 97.02% but only 83.08% on the test set, indicating potential overfitting. Similarly, XGBoost showed a training accuracy of 97.12% and a testing accuracy of 86.07%. These discrepancies highlight the importance of robust evaluation strategies like cross-validation to improve generalizability.

Table 2 presents the detailed performance metrics after hyperparameter tuning. Among all models, XGBoost outperformed the others, achieving the highest testing accuracy (86.07%) and F1-score (0.88), while maintaining a good balance between training and testing performance, suggesting lower overfitting.

Among the models tested, XGBoost delivered the most consistent and reliable performance. This can be attributed to several practical strengths of the algorithm. For instance, XGBoost includes built-in regularization techniques (both L1 and L2), which help to reduce overfittingan issue we noticed with some of the other models. It also handles missing values internally and works well with imbalanced datasets, which is particularly useful when dealing with real-world medical data like ours, where such imbalance is common.

In addition, using Stratified fivefold Cross-Validation made a significant difference in improving generalization across all models. We observed that some models initially showed very high training accuracy but much lower performance on the test set, indicating overfitting. Applying cross-validation helped to narrow this gap by validating the models on multiple subsets of the data, leading to a more balanced and trustworthy performance. This also increased our confidence in selecting XGBoost as the most effective model for this problem. Nevertheless, to further strengthen generalizability, external validation using independent datasets from other hospitals will be pursued in future work.

Feature Importance and Interpretability

To better understand how the model makes predictions, we examined feature importance using SHAP (SHapley Additive Explanations). This method offers insights into how individual features influence the model's decisions for each prediction. Figure 12 shows the SHAP summary plot generated for the best performing model, which is XGBoost. In this plot, the features are ranked by their overall impact on the prediction results. Each point in the plot represents one patient. The horizontal position of a point shows how much that particular feature contributed to the prediction of

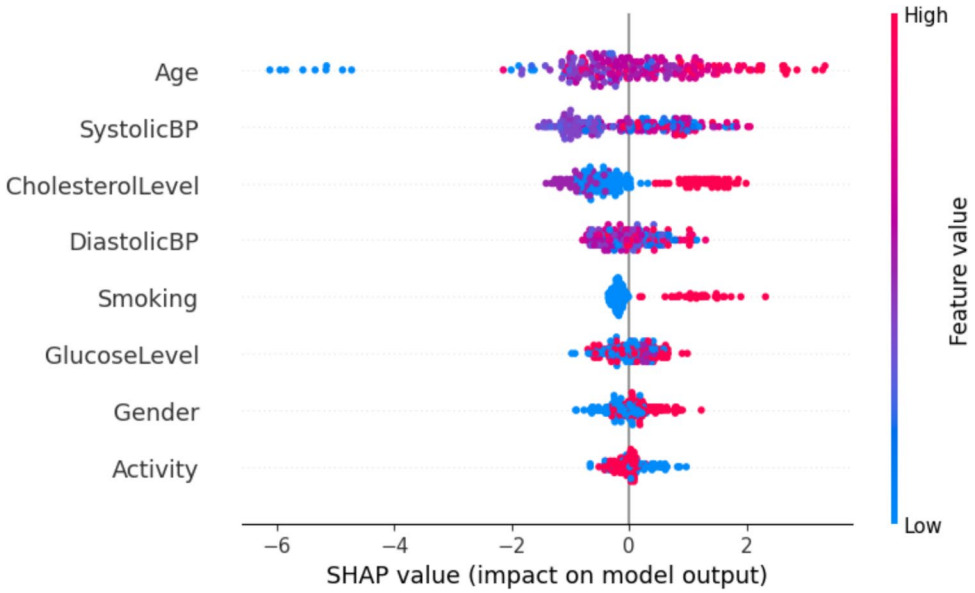


Fig. 12 SHAP summary plot for the XGBoost model, showing the relative influence of each feature on CVD predictions across all patients

cardiovascular disease for that person. Points appearing further to the right have a stronger positive influence on predicting the presence of disease. Those on the left tend to reduce the likelihood of disease prediction. The color of the points shows whether the feature value was high or low for that patient. Red means the value of that feature was high and blue means the value was low. For instance, we can see that for the Age feature, the red points are mostly located on the right. This means that older patients have a stronger influence on predicting cardiovascular disease. The same pattern is visible for Systolic Blood Pressure and Cholesterol Level, where higher values also increase the likelihood of disease. These patterns match well with known medical knowledge, where older age, high blood pressure, and high cholesterol are risk factors. In contrast, features like Physical Activity show a different trend. Here, higher values which indicate active individuals are mostly blue and appear on the left. This means that physical activity helps reduce the chance of being predicted as having the disease.

We applied only global SHAP analysis to highlight overall feature contributions across the dataset. Local patient level SHAP explanations were not included because the study is not explicitly designed as an XAI focused work. By focusing on global analysis, we were able to identify consistent patterns of feature influence that align with established cardiovascular risk factors thereby strengthening confidence in the model's clinical relevance. This clarification ensures that our findings are interpreted as population level feature importance rather than case-specific justifications. For example, global SHAP indicates that age and cholesterol are strong predictors of CVD risk on average across all patients, but it does not explain why the model predicts high risk for a specific individual with a unique combination of attributes.

This analysis not only confirms the medical relevance of the important features but also improves the transparency of the model. It shows that the model is focusing on the right indicators and supports its potential use in real clinical environments.

Table 3 summarizes the statistical validation of SHAP-based feature importance. Age, systolic blood pressure, and cholesterol level emerged as the most influential predictors, each showing high mean SHAP values with narrow confidence intervals and highly significant p-values ($p < 0.001$). Other features such as diastolic blood pressure, smoking, and glucose level also demonstrated significant contributions, while gender and physical activity showed comparatively smaller but still statistically relevant effects.

We already know that an AUC close to 1 signifies a highly effective model with strong separation capabilities, making it adept at distinguishing between positive and negative instances. Conversely, an AUC nearing 0

Table 3 Statistical validation of SHAP feature importance using bootstrap confidence intervals and permutation tests

Feature	Mean |SHAP|	95% CI (2.5–97.5%)	Permutation p-value
Age	1.09	0.93–1.25	< 0.001
SystolicBP	0.85	0.78–0.92	< 0.001
CholesterolLevel	0.74	0.68–0.80	< 0.001
DiastolicBP	0.39	0.36–0.43	0.02
Smoking	0.37	0.31–0.43	< 0.001
GlucoseLevel	0.32	0.30–0.35	< 0.001
Gender	0.27	0.24–0.30	0.01
Activity	0.19	0.16–0.21	0.04

suggests a less effective model, indicating challenges in accurately separating classes. This concise assessment of AUC provides a quick and insightful summary of the model's discriminatory power in this study. In analysis of various classifiers, Random Forest and XGBoost stands out with the highest AUC of 0.92 and 0.91 which can be seen on Fig. 13. This AUC value reflects its superior ability to distinguish between positive and negative instances, making it the most effective classifiers among those evaluated in the study. Our findings that XGBoost achieved the highest predictive are consistent with prior studies. Kedia et al. [11] also identified XGBoost as the best classifier on Indian hospital data, while Khan et al. [13] reported Random Forest as superior in a Pakistani cohort. Nikam et al. [9] found Decision Trees improved by BMI inclusion, and [10] observed strong performance from KNN and Random Forest. Within Bangladesh, Chowdhury et al. [5] achieved higher accuracy with SVM (91%) on a smaller Sylhet dataset, whereas Fahim et al. [15] confirmed XGBoost's superiority (81%) on local patient data. These comparisons suggest that while performance varies with dataset size, features, and region, ensemble methods such as XGBoost consistently demonstrate robustness across diverse South Asian populations.

Conclusion and Future Work

This study aimed to predict cardiovascular disease (CVD) using machine learning models, leveraging patient data collected from two hospitals in Dhaka, Bangladesh, to enhance prediction accuracy. Stratified fivefold cross-validation was employed, ensuring balanced class distribution across folds, which is particularly valuable for imbalanced datasets. Performance evaluation metrics, including accuracy, precision, recall, F1 score, and ROC-AUC, highlighted XGBoost (XGB) as the most effective model. After cross-validation, XGB achieved a training accuracy of 97.12% and a testing accuracy of 86.07%, along with superior precision (0.89), recall (0.88), and F1 score (0.88). These findings suggest that XGBoost holds promise for CVD prediction, though its clinical utility should be interpreted with caution until validated on larger and more diverse populations.

Although this study utilized a traditional train-test holdout approach and internal cross-validation to mitigate overfitting, external validation across independent hospital datasets was not performed due to access limitations. Incorporating such cross-hospital validation in future work will improve model generalizability and strengthen the clinical applicability of the results. In addition, the dataset used in this study exhibited a moderate imbalance between positive and negative CVD cases. While stratified cross-validation helped preserve class ratios during evaluation, more

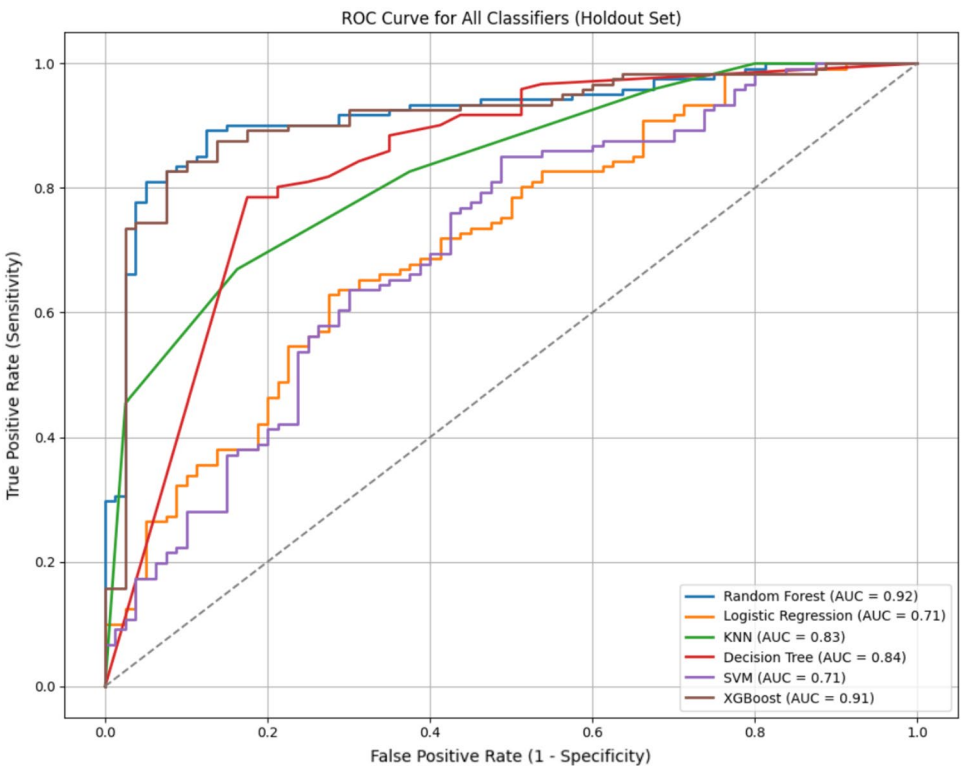


Fig. 13 Comparative analysis of the ROC curve produced by several classifiers

advanced imbalance handling techniques were not applied. Future work will explore these methods to ensure balanced performance across both classes.

We also recognize the potential of deep learning models such as Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs) in capturing complex, non-linear relationships in medical data. CNNs could be beneficial if image-based or sequential health records are integrated into the dataset. These architectures can automatically learn hierarchical patterns that may be missed by traditional models. Given the increasing availability of electronic health records (EHR) and wearable sensor data, future studies will explore such architectures to enhance predictive accuracy further.

Additionally, considering prior findings like those from Khan et al. (2018) regarding obesity's role in CVD, future datasets will include BMI and other lifestyle indicators that were unavailable in the current study. This expansion will help build a more comprehensive and generalizable model capable of supporting real-world clinical decision-making.

Furthermore, while this study focused on overall SHAP analysis to highlight the key features driving predictions across the dataset future work will extend interpretability methods to provide more individualized and patient specific insights that can support clinical decision making.

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Data Availability The dataset supporting the findings of this study cannot be shared publicly due to patient confidentiality and institutional restrictions. Data were collected from the National Institute of Cardiovascular Diseases and the Ibrahim Cardiac Hospital and Research Institute in Dhaka, Bangladesh. Qualified researchers may request access to the anonymized dataset by submitting a formal proposal to the respective hospital authorities. Access will only be granted following approval from the institutional ethics committee and the completion of a data-sharing agreement in compliance with confidentiality requirements.

Code Availability None.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval This study was conducted in compliance with ethical guidelines and approved by the institutional ethics committee (both "National Institute of Cardiovascular Diseases" and the "Ibrahim Cardiac Hospital and Research Institute").

Informed Consent Informed consent was obtained from all participants involved in this study.

Consent for Publication All authors have reviewed and approved the manuscript and consent to its publication.

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